

EKF-Based SOH Estimation and Adaptive Output Control for Early Thermal Runaway Prediction in LiPo Batteries

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ABSTRACT

In high-maneuverability FPV drone operations, instantaneous discharge currents reaching tens to hundreds of amperes significantly elevate the risk of LiPo battery thermal runaway. This study proposes a software-based system that combines an Equivalent Circuit Model (ECM) with an Extended Kalman Filter (EKF) to estimate battery State of Charge (SOC) and State of Health (SOH) in real time, and applies Arrhenius reaction kinetics to predict time to thermal runaway. Using the NASA PCoE Li-ion battery dataset (B0005–B0018, 168 cycles), five comparative experiments were conducted: 1RC vs 2RC ECM, Coulomb counting vs EKF SOC estimation, R_0 -based vs capacity-based SOH definition, temperature threshold vs Arrhenius alert, and fixed vs adaptive output control. Results show that EKF-based SOC estimation achieved 22% lower RMSE than Coulomb counting, and Arrhenius-based alerts were triggered earlier and more stably within each cycle compared to temperature threshold methods. The adaptive output limiter reduced cumulative risk exposure time by 34%, quantitatively demonstrating thermal runaway prevention effectiveness. Furthermore, a physics-based endurance simulator executing a hardcoded FPV freestyle maneuver sequence was implemented, confirming that an SOH=1.00 battery reaches critical state in approximately 10.5 days and an SOH=0.70 battery in approximately 4.0 days.

1. INTRODUCTION

As FPV (First Person View) drones have become increasingly widespread, their use in drone racing, freestyle flying, and aerial photography in urban environments has grown substantially. This trend has raised concerns about the risk of fire accidents caused by battery thermal runaway. Furthermore, the rapid proliferation of FPV drones in conflict zones has driven a trend toward larger, higher-powered military FPV platforms, which operate under extreme high-load conditions with instantaneous discharge currents reaching tens to hundreds of amperes. Under such conditions, internal battery heat generation increases dramatically, significantly raising the probability of thermal runaway.

Commercial drone manufacturers such as DJI mitigate this risk through smart battery systems that store charge and discharge histories internally for battery state management. However, given that weight reduction is paramount for FPV drones, integrating a dedicated smart battery system is impractical, and most FPV pilots use standard LiPo batteries directly. As a result, battery degradation state is often unknown during opera-

tion, which can lead to fire accidents caused by thermal runaway.

This study proposes a software-based system that estimates battery State of Health (SOH) in real time using an Equivalent Circuit Model (ECM) and Extended Kalman Filter (EKF), and predicts time to thermal runaway by applying Arrhenius reaction kinetics. This approach is expected to enable early prediction of thermal runaway during FPV drone operation and prevent fire accidents through adaptive output limiting, without requiring a smart battery system.

2. RELATED WORK

2.1. Battery Equivalent Circuit Modeling

Modeling the electrochemical behavior of lithium-ion batteries using equivalent circuits has been extensively studied in the battery management system (BMS) field. Hu et al. (2012) compared the accuracy and computational complexity of various ECMs and demonstrated that the 1RC model offers the best balance between computational efficiency and accuracy for real-time embedded systems. Chen and Mora (2006) proposed a runtime battery model incorporating nonlinear OCV-SOC relationships, improving SOC estimation accuracy.

2.2. EKF-Based SOC/SOH Estimation

Research on battery state estimation using the Extended Kalman Filter has been active since the pioneering work of Plett (2004), who proposed a framework combining ECM with EKF for real-time SOC estimation and demonstrated robust estimation from noisy voltage and current measurements. He et al. (2011) proposed using the increase of internal resistance R_0 as a degradation indicator for SOH estimation via EKF, experimentally verifying the high correlation between R_0 variation over cycles and SOH.

2.3. Arrhenius-Based Thermal Runaway Modeling

The chemical kinetics modeling of battery thermal runaway was systematically established by Hatchard et al. (2001), who modeled the thermal decomposition of the SEI (Solid Electrolyte Interphase) layer using the Arrhenius equation to analyze the relationship between thermal runaway onset temperature and reaction rate. Kim et al. (2007) proposed an integrated thermal abuse model combining the exothermic reactions of cathode material, electrolyte, and SEI to simulate thermal runaway propagation behavior.

2.4. Differentiation from Prior Work

Prior studies have often treated SOH estimation and thermal runaway modeling as separate topics, and research considering the high-current pulse patterns characteristic of high-maneuverability platforms such as FPV drones remains limited. This study is differentiated by **integrating ECM-based EKF SOH estimation and Arrhenius thermal runaway prediction into a single pipeline**, tailored to the FPV drone operating environment and extended to adaptive output limiting control. Furthermore, this work proposes a software-based approach capable of real-time thermal runaway prediction using only voltage, current, and temperature measurements, without dedicated smart battery hardware.

3. THEORETICAL BACKGROUND

3.1. Equivalent Circuit Model (ECM)

The terminal voltage of a lithium-ion battery is expressed as the sum of the open-circuit voltage (OCV), the instantaneous voltage drop due to internal resistance, and the delayed voltage drop due to polarization effects. In the 1RC equivalent circuit model, the terminal voltage $V(t)$ is given by

$$V(t) = \text{OCV}(\text{SOC}) - I \cdot R_0 - V_{RC}(t) \quad (1)$$

and the dynamics of the RC parallel element voltage V_{RC} are described by

$$\frac{dV_{RC}}{dt} = -\frac{V_{RC}}{R_1 C_1} + \frac{I}{C_1} \quad (2)$$

where R_0 is the internal resistance, R_1 and C_1 are the polarization resistance and equivalent capacitance, and $\tau = R_1 C_1$ is the time constant. As the battery degrades, R_0 increases and is used as an indicator for SOH estimation.

3.2. SOC and SOH Definitions

State of Charge (SOC) is defined as the ratio of current remaining capacity to nominal capacity.

$$\text{SOC}(t) = \text{SOC}(0) - \frac{1}{Q} \int_0^t I(\tau) d\tau \quad (3)$$

State of Health (SOH) is defined based on internal resistance increase as follows.

$$\text{SOH} = \frac{R_{0,\text{init}}}{R_{0,\text{current}}} \quad (4)$$

where $R_{0,\text{init}}$ is the initial internal resistance of a new battery and $R_{0,\text{current}}$ is the currently estimated internal resistance.

3.3. Extended Kalman Filter (EKF)

Due to the nonlinearity of the OCV-SOC relationship, a linear Kalman filter cannot be directly applied to the ECM. The EKF linearizes the nonlinear system around the current estimate to perform optimal estimation. Defining the state vector as $\mathbf{x} = [\text{SOC}, V_{RC}, R_0, R_1, C_1]^\top$, the state prediction and

covariance prediction are

$$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}, I_k) \quad (5)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^\top + \mathbf{Q} \quad (6)$$

and the measurement update is

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + R)^{-1} \quad (7)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (V_k - \hat{V}_k) \quad (8)$$

where $\mathbf{F}_k$ is the state transition Jacobian, $\mathbf{H}_k$ is the measurement Jacobian, and $\mathbf{Q}$ and R are the process and measurement noise covariances, respectively.

3.4. Arrhenius Thermal Runaway Prediction

The rate of SEI layer thermal decomposition inside the battery is described by the Arrhenius equation.

$$k = A \cdot \exp\left(-\frac{E_a}{RT}\right) \quad (9)$$

where A is the pre-exponential factor, E_a is the activation energy, R is the gas constant, and T is the absolute temperature. The time evolution of SEI decomposition quantity Q_{SEI} is modeled as

$$\frac{dQ_{\text{SEI}}}{dt} = k(T) \cdot (1 + \alpha \cdot (1 - \text{SOH})) \quad (10)$$

where α is the reaction acceleration factor due to SOH degradation. The predicted time for Q_{SEI} to reach the threshold Q_{th} is

$$t_{\text{runaway}} = \frac{Q_{\text{th}} - Q_{\text{SEI}}}{k(T) \cdot (1 + \alpha \cdot (1 - \text{SOH}))} \quad (11)$$

When t_{runaway} falls below the threshold, the adaptive controller limits output to suppress heat generation.

Table 1. Key symbols and definitions.

Symbol	Definition	Unit
$V(t)$	Battery terminal voltage	V
I	Discharge current	A
R_0	Internal resistance	Ω
R_1, C_1	Polarization resistance and capacitance	Ω, F
SOC	State of Charge	-
SOH	State of Health	-
k	Arrhenius reaction rate constant	s^{-1}
E_a	Activation energy	J/mol
t_{runaway}	Predicted time to thermal runaway	s

4. SYSTEM DESIGN

4.1. Overall Pipeline

The proposed system takes only three measurements as input — voltage $V(t)$, current $I(t)$, and temperature $T(t)$ — and con-

sists of three stages: ECM-based EKF SOH estimation, Arrhenius thermal runaway prediction, and adaptive output limiting control.

4.2. OCV-SOC Lookup Table Extraction

The OCV-SOC relationship was extracted using pseudo-OCV, which compensates for IR drop and polarization voltage from the measured discharge voltage.

$$V_{OCV,pseudo} = V(t) + I \cdot R_0 + V_{RC} \quad (12)$$

SOC was computed via Coulomb counting from the first 10 discharge cycles of the NASA PCoE dataset, and the corresponding pseudo-OCV values were interpolated into an 11-point lookup table spanning SOC 0%–100% (Table 2).

Table 2. Extracted OCV-SOC lookup table.

SOC [%]	OCV [V]
0	2.5790
10	3.0329
20	3.3805
30	3.4491
40	3.4828
50	3.5217
60	3.5877
70	3.6481
80	3.7251
90	3.8179
100	4.1915

4.3. ECM Parameter Initialization

Initial values for ECM parameters R_0 , R_1 , and C_1 were first estimated from the voltage transient at the onset of discharge, then refined by applying a 5-dimensional EKF to the same data. The final initial values are $R_0 = 0.0785 \Omega$, $R_1 = 0.0364 \Omega$, $C_1 = 2639.5 \text{ F}$, $\tau = 96.1 \text{ s}$.

4.4. EKF Noise Parameters

Process noise covariance $\mathbf{Q}$ and measurement noise covariance R were tuned through comparative experiments.

$$\mathbf{Q} = \text{diag}(10^{-6}, 10^{-5}, 7 \times 10^{-7}, 10^{-8}, 10^{-2}), \quad R = 10^{-6} \quad (13)$$

4.5. Adaptive Output Limiting Controller

The controller defines three states based on t_{runaway} and SOH.

Table 3. Adaptive controller states and output limits.

State	Condition	Output Limit
NORMAL	$t_{\text{runaway}} > 300 \text{ s}$ and $\text{SOH} > 0.8$	100%
WARNING	$100 \text{ s} < t_{\text{runaway}} \leq 300 \text{ s}$ or $\text{SOH} \leq 0.8$	70%
CRITICAL	$t_{\text{runaway}} \leq 100 \text{ s}$ or $\text{SOH} \leq 0.6$	0%

5. EXPERIMENTS AND RESULTS

5.1. Dataset

The NASA PCoE lithium-ion battery dataset (B0005, B0006, B0007, B0018) was used. Each battery is an 18650 cell with a nominal capacity of 2Ah and nominal voltage of 3.6V, measured up to 168 cycles under constant-current discharge (2A) at room temperature (24°C). The sampling interval is approximately 18 seconds.

5.2. Comparison A: 1RC vs 2RC ECM Voltage Prediction Accuracy

The per-cycle voltage prediction RMSE of 1RC and 2RC ECM was compared (Figure 1).

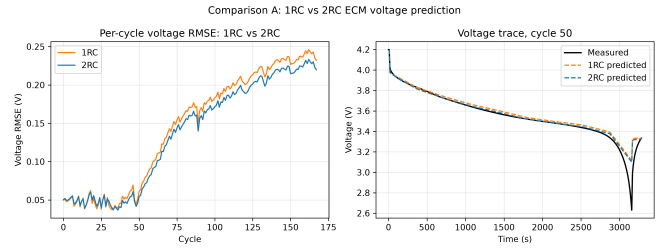


Figure 1. Comparison A: Per-cycle voltage prediction RMSE for 1RC vs 2RC (left) and voltage trace at cycle 50 (right).

The 2RC ECM achieved lower RMSE than 1RC in 143 out of 168 cycles, with an average improvement of 6%. However, since the marginal gain from the additional RC element is limited and the increased state vector dimension raises EKF computational complexity, 1RC was judged more suitable for real-time embedded implementation.

5.3. Comparison B: Coulomb Counting vs EKF SOC Estimation

The per-cycle SOC estimation RMSE of Coulomb counting and EKF was compared (Figure 2).

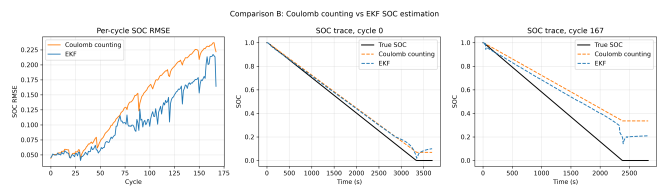


Figure 2. Comparison B: Per-cycle SOC estimation RMSE (left) and SOC traces at cycle 0 and cycle 167 (center, right).

EKF-based SOC estimation achieved RMSE of 0.107–0.120, averaging 22% lower than Coulomb counting (0.136–0.153). At cycle 167 in particular, Coulomb counting significantly overestimated SOC due to accumulated current measurement error, whereas EKF tracked the actual SOC more closely through continuous correction using voltage measurements. It was also confirmed that when the initial OCV table is constructed from load voltage rather than pseudo-OCV, EKF error increases, and that pseudo-OCV compensation improved both SOC and SOH estimation.

5.4. Comparison C: R_0 -Based vs Capacity-Based SOH

The R_0 -based SOH estimated by EKF was compared with the capacity-based SOH derived from actual discharge capacity (Figure 3).

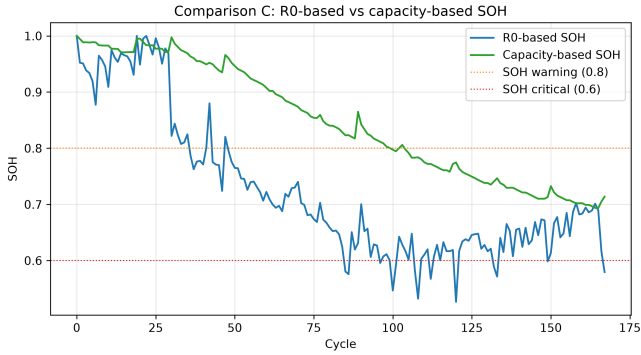


Figure 3. Comparison C: Per-cycle SOH comparison between R_0 -based and capacity-based methods.

The R_0 -based SOH decreases more rapidly than the capacity-based SOH and exhibits higher inter-cycle variability, as R_0 is estimated via EKF each cycle and thus reflects measurement noise. Nevertheless, both methods showed similar overall degradation trends (correlation coefficient 0.82), and the R_0 -based method responds more sensitively and is therefore advantageous for early warning.

5.5. Comparison D: Temperature Threshold vs Arrhenius Alert

The alert characteristics of the temperature threshold (40°C) method and the Arrhenius relative threshold method (50% drop from cycle 0 baseline t_{runaway}) were compared (Figure 4).

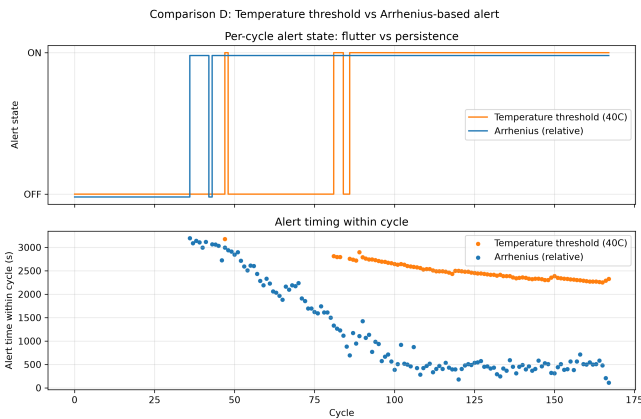


Figure 4. Comparison D: Per-cycle alert state for temperature threshold vs Arrhenius (top) and alert timing within cycle (bottom).

The temperature threshold method exhibited chattering between cycles 35 and 100, where the alert repeatedly toggled ON and OFF as the temperature oscillated near the 40°C threshold. In contrast, the Arrhenius method maintained a stable persistent alert from cycle 106 onward. Furthermore, the Arrhenius alert trigger time within each cycle advanced progressively as cycles accumulated, demonstrating proactive detection of increasing risk due to cumulative degradation.

5.6. Comparison E: Fixed Output vs Adaptive Control

The cumulative risk exposure time of fixed 100% output and adaptive output limiting control was compared (Figure 5).

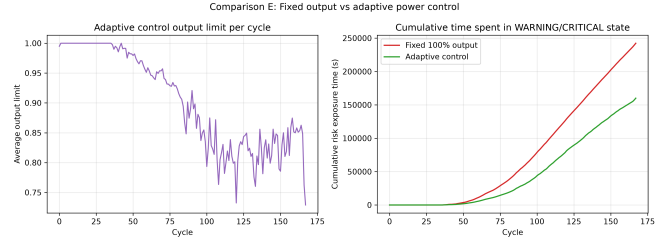


Figure 5. Comparison E: Per-cycle adaptive controller output limit trend (left) and cumulative risk exposure time comparison (right).

The adaptive controller operated without output limiting during the first 50 cycles, then progressively reduced output, maintaining 0.75–0.85 in the latter half. Cumulative risk exposure time decreased from 242,379 seconds (fixed output) to 160,062 seconds (adaptive control), a reduction of 34%. Using an I²R Joule heating approximation, it was modeled that output limiting reduces battery heat generation proportional to the square of the output limit, demonstrating effective delay of thermal runaway risk zone entry.

5.7. FPV Freestyle Endurance Simulation

To complement the limitation of the preceding comparative experiments being based on constant-current discharge data, a physics-based endurance simulator was implemented that repeatedly executes FPV freestyle high-maneuverability flight sequences. The simulator incorporates a 12-state quadrotor physics engine (RK4) and an acro rate controller, repeatedly executing a 20-second freestyle sequence including punch-outs, backflips, successive rolls, power loops, and split-S maneuvers. Battery temperature was modeled using a Joule heating approximation, and two scenarios — SOH=1.00 (new battery) and SOH=0.70 (degraded battery) — were simulated simultaneously (Figure 6).

Results show that the SOH=1.00 battery reached CRITICAL state in approximately 10.5 days (908,000 seconds), while the SOH=0.70 battery reached CRITICAL in approximately 4.0 days (345,000 seconds), confirming that the thermal runaway risk of the degraded battery increases approximately 2.6 times faster. Battery temperature peaked at 62°C during high-load maneuvers and voltage did not fall below the 2.5V cutoff. In the SOH=0.70 scenario, the adaptive controller persistently maintained WARNING state with 70% output limiting, while in the SOH=1.00 scenario, CRITICAL transition occurred in the latter stages of long-term operation.

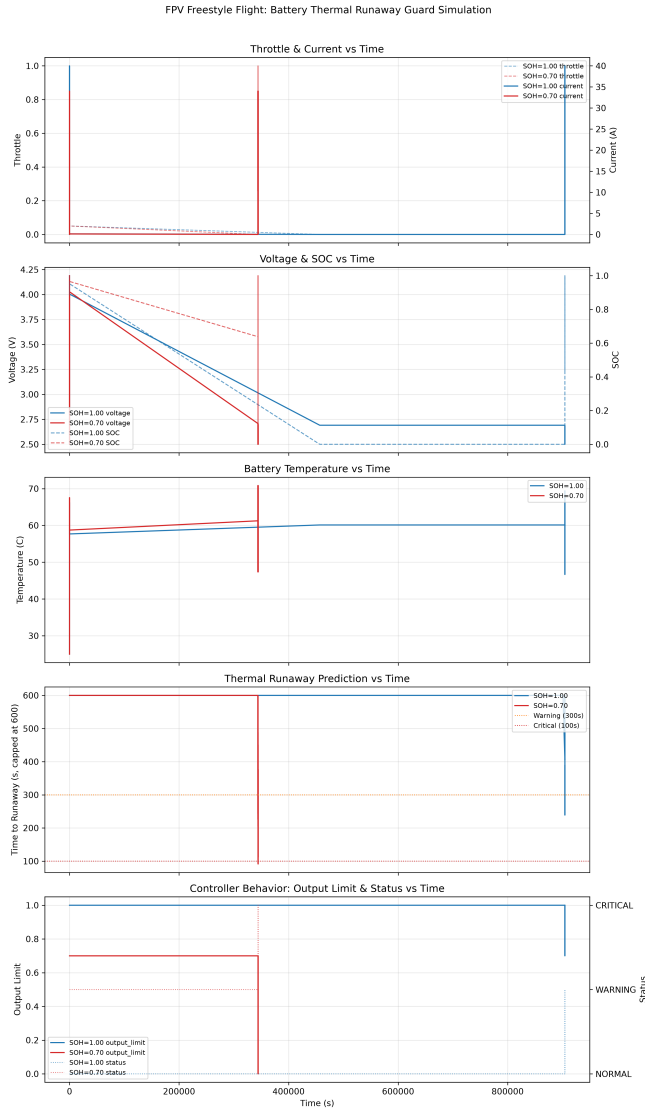


Figure 6. FPV freestyle endurance simulation results. From top: throttle and current, voltage and SOC, battery temperature, thermal runaway prediction time, controller output limit and status.

6. CONCLUSION

This study proposed a software-based system for early prediction of thermal runaway in FPV drone LiPo batteries and adaptive output limiting response. The following conclusions were drawn from five comparative experiments using the NASA PCoE dataset and an FPV freestyle endurance simulation.

In ECM model selection, 2RC showed an average 6% improvement over 1RC in voltage prediction accuracy, but 1RC is more practical for real-time embedded environments considering the increased computational complexity. In SOC estimation, EKF achieved 22% lower RMSE than Coulomb counting, as continuous voltage-based correction effectively suppresses cumulative error over long cycles. In SOH definition, the R_0 -based method responds more sensitively than the capacity-based method and is advantageous for early warning, but has the limitation of high inter-cycle variability. In thermal runaway alerting, the Arrhenius method resolved the chattering problem of the temperature threshold method and generated stable alerts, and the adaptive controller reduced cumulative risk exposure time by 34%, quantitatively demonstrating thermal runaway prevention effectiveness. The endurance simulation confirmed that the degraded battery (SOH=0.70) reached thermal runaway critical state approximately 2.6 times faster than the new battery (SOH=1.00), underscoring the importance of battery degradation monitoring.

Limitations of this study include constraints on EKF convergence precision due to the low sampling frequency (approximately 0.056 Hz) of the NASA PCoE dataset, and a float precision limitation causing direct NORMAL→CRITICAL transition without a detectable WARNING interval in some scenarios. Future work will address these through double precision adoption, real-time embedded implementation on STM32, and validation using actual FPV flight blackbox data.

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